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PERSONIFIED TEST FOR PROFESSIONAL SUITABILITY FOR WORKING IN EXTREME NORTHERN CONDITIONS

L.Y. Zakharinskiy[1]
Scientific supervisor L.V. Stepanova[1]
Candidate of Biological Sciences, Associate Professor

[1]*Siberian Federal University*

Introduction. Currently, many states are interested in the development of the Arctic as the region with high potential.

Transport accessibility is one of the key issues of the North. Aviation is the only reliable year-round mode of transport for delivering people and goods to the Arctic. Therefore, pilots and air traffic controllers are among the most promising and in-demand specialists in the Arctic regions.

However, Polar aviation faces a constant shortage of qualified personnel. More than 70% of specialists leave the profession within 20 years after graduation from their training programs.

One of the reasons is the physiological inability of the body to adapt to prolonged voluntary, emotional, and intellectual stress under the extreme conditions of the Far North [1].

A solution to the problem could be the development of a biotest that allows identifying people with outstanding adaptive abilities of the body for further work in the Arctic.

Changes in the functional state of the body (caused by stress) can be assessed through saliva, as it is a very dynamic biological fluid.

As a method, we propose bioluminescent testing of saliva, which allows assessing the physiological state of a person based on the level of luminescence of the enzymatic reaction [2].

Scientific novelty: for the first time, the possibility of using bioluminescent analysis of saliva to assess professional suitability has been demonstrated.

The data obtained can be used as a tool for professional overloads selection and prevention.

The goal of the research: to identify the possibility of using bioluminescent testing of saliva for professional selection operation.

Hypothesis: the adaptive abilities of the organism can be assessed by the change in the bioluminescent indicator of saliva.

Materials and methods of research. Young men aged 18-20 years who passed the selection of the medical-flight expert commission and are studying at the Krasnoyarsk

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branch of the St. Petersburg State University of Civil Aviation. Their specialties are a pilot and air traffic controller whose course of study lasts over a period of 5 days.

Testing was conducted during the exam period (December 2023) on core subjects, i.e., the experiment participants experienced peak stress loads during those days.

At the test beginning, written informed consent for data processing and voluntary participation in the experiment was obtained from the participants.

To identify factors conditionally affecting the analysis, a self-assessment questionnaire was conducted, including data on socio-demographic aspects, lifestyle, and health indicators.

The material of the study was saliva, which was collected by spitting into a sterile tube before and after the load (7:00 AM and 5:00 PM accordingly).

Before testing, the saliva was centrifuged for 15 minutes at 5000 rpm. The supernatant was used for analysis.

Bioluminescent testing of saliva was conducted using a multi-component immobilized reagent "Enzymolum" (LLC "NPP Applied Biosystems"). The catalyzing substrate was 0.16 mM aqueous solution of flavin mononucleotide (FMN) (Serva, Germany). The maximum intensity of luminescence was recorded on a "Glomax 20/20" luminometer (Promega, USA), according to the developed methodology [2]. The reaction of biotests was determined by the magnitude of the bioluminescent indicator (T, %).

To identify the stress resistance of the organism, the difference in the values of the bioluminescent indicator before and after the load ($\Delta T = T_{after} - T_{before}$) was used.

The concentration of cortisol in saliva was determined by an enzyme-linked immunosorbent assay by the staff of the Research Institute of Medical Problems of the North, Krasnoyarsk. Indicators exceeding the reference values indicated the stress presence in the body [1].

Statistical data processing was carried out by calculating the mean and standard deviation. Data analysis was performed in Visual Studio Code (Microsoft, USA), using machine learning and data analysis libraries.

Research results are the following: 60% of the subjects had a normal body mass index (BMI), 20% had its deficit, and 20% had pre-obesity. Participants with a normal BMI regularly engaged in sports, while those with a deviant BMI had harmful habits, adhered to diets, and took medications.

Bioluminescent testing of saliva over 5 days showed that ΔT changed according to a specific pattern for each group of participants (table).

To identify stress levels, cortisol concentrations before and after the load (ΔCor) were analyzed over 5 days for each group of experiment participants (table).

Table

Summary Analysis of Data on ΔT and ΔCor

No. Gr	ΔT, %	ΔCor, nmol/L	%	Adaptive Abilities
1	Minor changes	Changes within normal limits (reference values)	20	Outstanding
2	Stable decrease of a minor magnitude	Minor deviations from the norm after load	40	Good
3	Stable increase, significant magnitude	Borderline normal values before load and minor deviations after load	30	Satisfactory
4	Significant fluctuations (increases or decreases)	Significant exceedances of the norm, either before or after the load	10	Low

Correlational analysis confirmed the relationship between ΔT and ΔCor. The four groups of experiment participants (table) can be associated with four levels of stress resistance of the organism [3].

A comprehensive analysis of the health status data of participants revealed that young men with outstanding adaptive abilities lead a healthy lifestyle, while those with satisfactory indicators have the potential to strengthen their health in case, they give up harmful habits.

Conclusion. The bioluminescent saliva analysis allows to provide the organism's stress resistance with an integrated assessment.

Based on changes in the bioluminescent indicator, in correlation with cortisol concentration, groups with different levels of adaptive ability were identified.

The innovative, personalized, non-invasive, accessible, and easily scalable bioluminescent biotest can be considered promising for assessing the stress state of the organism. With further development, it can be applied not only in Polar aviation but also in other industries that require professional selection.

References

1. Khasnulin V. I., Khasnulina A. V. Characteristics of psycho-emotional stress among residents of Northern and Siberian regions with uncomfortable climate conditions at high and low levels of stress hormones in the blood // World of Science, Culture, Education. 2012. No. 5. P. 32-35.

2. Esimbekova E. N., Torgashina I. G., Kalyabina V. P., Kratasyuk V. A. Enzymatic biotesting: scientific foundations and applications // Siberian Environmental Journal. 2021. Vol. 14, No. 3. P. 290-304.

3. Four groups of stress resilience. Socionics and Psychosophy, 28.01.2020. [Electronic resource]. URL: https://www.socionika.lv/ru/отношения-и-стресс/четыре-группы-стресса-устойчивости/ [accessed 13.03.2024].